UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering

A Hard-Voting Ensemble Approach for Cyberbullying Detection

Technical Report

Md. Fahim Ahmed
ID: 0112330413
Group: 10

Course: Machine Learning (CSE 4889)
Section: F
Course Teacher: Kazi Fahim Tahmid

Abstract

Cyberbullying on social media is too big for manual moderation and keyword filters normally fail on slang and context. This report present a feature-enriched hard-voting ensemble that classifies tweets as cyberbullying or not-cyberbullying. Each tweet is cleaned and represented by 8,000 TF-IDF n-gram features (unigrams to trigrams) concatenated with 11 scaled linguistic, VADER sentiment and offensive-word features, giving 8,011 features. Five classifiers (LinearSVC, Logistic Regression, Random Forest, Multinomial Naïve Bayes and KNN) are combined by majority voting. On a balanced subset of 17,805 tweets from the Kaggle Cyberbullying Classification dataset, the ensemble reaches 83.08% test accuracy with F1 = 0.84 for the cyberbullying class and 82.75% mean 5-fold cross-validation accuracy. All models are compared under the same data, features and splits. The ensemble outperforms SVM, Naïve Bayes and KNN, is on par with Logistic Regression, and is about 0.7 points below the best single model, Random Forest (83.8%). Its main weakness is 288 missed cyberbullying tweets (false negatives).

Keywords: cyberbullying detection, ensemble learning, hard voting, TF-IDF, VADER, text classification

1. Introduction

1.1 Background and Problem Statement

Cyberbullying has become a significant problem on social media platforms, where users can easily post harmful, offensive, threatening or abusive content. The volume of user-generated content makes it difficult for human moderators to identify and respond to such messages manually. Traditional keyword-based approaches are often insufficient because cyberbullying can be expressed through many different words, phrases, contexts and writing styles.

A further challenge is that individual machine learning classifiers behave differently depending on the characteristics of the text: a model that performs well on one kind of text may perform

Github Repository: https://github.com/FahimAhmed121/cyberbullying-detection-hard-voting-ensemble

less effectively on another. There is therefore a need for an automated system that can analyze textual content and reliably distinguish between cyberbullying and non-cyberbullying content.

1.2 Motivation

Cyberbullying can negatively affect users' psychological well-being, social interactions and participation in online communities. An automated detector can help identify potentially harmful content more efficiently. This work is motivated by the idea that combining complementary information sources may be more dependable than relying on a single model:

- **TF-IDF features** capture important words and phrases.
- **Linguistic features** capture writing characteristics such as word count, average word length, capitalization and punctuation.
- **Sentiment features** capture positive, negative, neutral and overall emotional tone.
- **Multiple classifiers and hard voting** combine the decisions of SVM, Logistic Regression, Random Forest, Naïve Bayes and KNN.

1.3 Research Question and Objectives

Research question. Can a feature-enriched hard-voting ensemble that combines TF-IDF, linguistic and sentiment features provide more reliable cyberbullying detection than individual traditional machine learning classifiers?

The objectives are: (i) to build a unified preprocessing and feature pipeline producing 8,011 features per tweet; (ii) to train five traditional classifiers and a hard-voting ensemble on identical data; (iii) to evaluate all models using accuracy, precision, recall, F1-score, the confusion matrix and 5-fold cross-validation; and (iv) to provide a console-based live prediction tool.

1.4 Expected Benefit

The proposed approach aims to offer a relatively simple and computationally manageable ensemble framework for automated detection of harmful social-media content. If the ensemble shows improved or competitive performance compared with individual classifiers, it provides evidence that combining different feature types and classifier decisions is useful for this task.

2. Literature Review

Ensemble learning has been widely applied to cyberbullying detection. Three recent studies are most relevant to this work.

Alam, Bhowmik and Prosun (2021) [1] proposed an ensemble-based approach for cyberbullying detection on Twitter. They compared four individual classifiers (Multinomial Naïve Bayes, Logistic Regression, Decision Tree and Linear SVC) with ensemble strategies such as Gradient Boosting, AdaBoost and Bagging, using Bag-of-Words and TF-IDF with

different n-gram settings. Their single-level and double-level voting ensembles outperformed the individual classifiers, with a reported best accuracy of about 96%.

Alqahtani and Ilyas (2024) [2] built a stacking ensemble of Decision Tree, Random Forest, Linear SVC, Logistic Regression and KNN for detecting aggressive and non-aggressive tweets, testing Bag-of-Words, TF-IDF, Word2Vec and GloVe features. The stacking classifier reached about 94% accuracy, although stacking adds a meta-learning layer and extra complexity.

Singh, Metta and Patnaik (2025) [3] evaluated Naïve Bayes, SVM, Decision Tree, Random Forest, KNN and Gradient Boosting on TF-IDF features and combined them in a stacking ensemble with a Logistic Regression meta-classifier tuned by grid search, reporting 90.18% accuracy.

Study	Year	Features	Models	Ensemble	Reported result
Alam et al.	2021	BoW, TF-IDF, n-grams	MNB, LR, DT, LSVC + ensembles	SLE / DLE voting	≈ 96%
Alqahtani & Ilyas	2024	BoW, TF-IDF, Word2Vec, GloVe	DT, RF, LSVC, LR, KNN	Stacking	≈ 94%
Singh et al.	2025	TF-IDF	NB, SVM, DT, RF, KNN, GB	Stacking + LR	90.18%
Proposed work	2026	TF-IDF + linguistic + sentiment + offensive	SVM, LR, RF, NB, KNN	Hard voting	83.08%

Table 1. Comparison of related studies with the proposed work. Results are as reported by the authors and are not directly comparable (different datasets, preprocessing and protocols).

2.1 Research Gap

These studies show that ensemble learning has already been applied to cyberbullying detection, so this work does not claim novelty for ensembles in general. The gap addressed is more specific:

- **Feature combination:** the reviewed ensembles mainly use text representations (TF-IDF, BoW, Word2Vec, GloVe); less emphasis is placed on combining TF-IDF with linguistic, sentiment and offensive-word indicators.
- **Ensemble strategy:** stacking approaches add a meta-classifier; it remains of interest whether a simple hard-voting ensemble without a meta-learner can be competitive.
- **Controlled comparison:** reported accuracies differ in datasets and protocols, so a like-for-like comparison of SVM, LR, RF, NB, KNN and the ensemble under the same data, preprocessing, features, split and metrics is more defensible than comparing against published numbers.

3. Dataset

3.1 Source and Structure

The experiments use the Kaggle Cyberbullying Classification dataset (cyberbullying_tweets.csv) [4], containing 47,692 tweets with two columns: the raw tweet text (string) and a cyberbullying_type label (categorical) with six classes. The original classes are almost perfectly balanced, as shown in Table 2.

Original label	Tweets	Share (%)
religion	7,998	16.8
age	7,992	16.8
gender	7,973	16.7
ethnicity	7,961	16.7
not_cyberbullying	7,945	16.7
other_cyberbullying	7,823	16.4
Total	47,692	100.0

Table 2. Original label distribution.

3.2 Binary Labelling and Balancing

The task is cast as binary classification: not_cyberbullying is mapped to 0 and the five remaining classes are merged into class 1 (Cyberbullying), which creates 39,747 positive versus 7,945 negative tweets. Rows with missing values and texts that become empty after cleaning were removed, leaving 7,805 non-cyberbullying tweets. The cyberbullying class was randomly down-sampled to 10,000 tweets (random_state = 42), giving a final set of N = 17,805 tweets with a 56.2% / 43.8% class ratio, which is handled by stratified splitting. Tweets are short, informal English text averaging about 136 characters.

Figure 1 confirms that the five cyberbullying categories remain evenly represented after sampling (each about 20%), so the positive class is not dominated by any single type of abuse. Figure 2 shows the resulting class sizes.

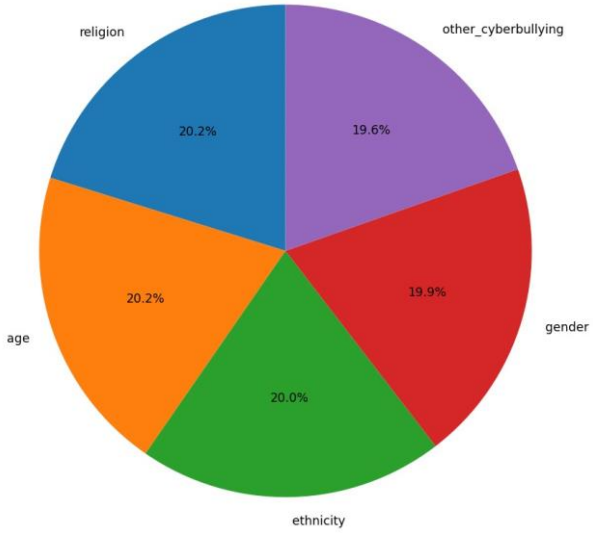


Figure 1. Share of the five cyberbullying categories after sampling.

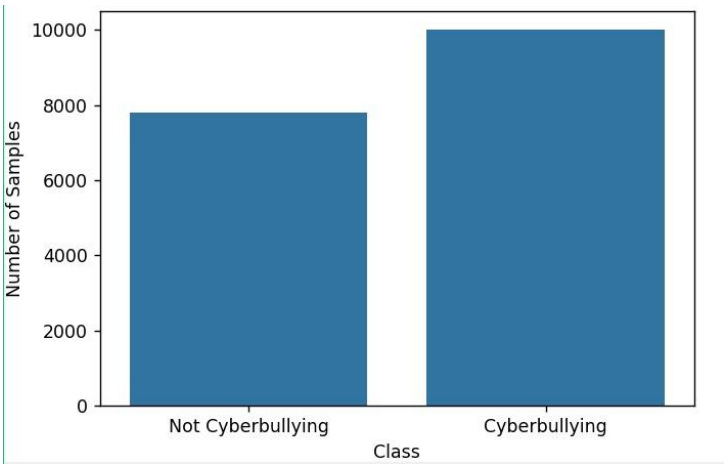


Figure 2. Class distribution after balancing (7,805 vs. 10,000).

4. Methodology

4.1 Overall Pipeline

The pipeline has five stages: (1) load the data and create binary labels; (2) preprocess text; (3) extract and combine features; (4) train five classifiers and the hard-voting ensemble; and (5) evaluate on held-out data with metrics and cross-validation (Figure 3). All models share the same preprocessing, features and data splits.

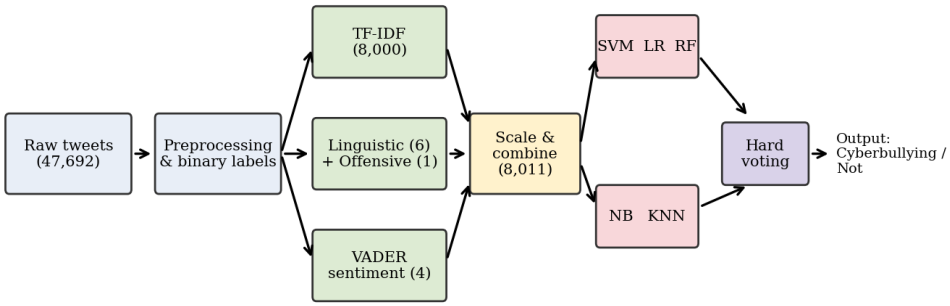


Figure 3. Overall pipeline of the proposed system.

4.2 Text Preprocessing

- Lowercasing and removal of URLs, @mentions and hashtags.
- Contraction expansion (can't → cannot, n't → not, 're → are, etc.).
- Repeated punctuation (!!, ??, ...) converted to marker tokens, then removal of punctuation and extra whitespace.
- Stopword removal using the NLTK English list while retaining negation words (not, no, never, nothing, nobody, nowhere, neither, nor), since negation can reverse the meaning of an insult.

4.3 Feature Engineering

Each tweet is represented by 8,011 features (Table 3). Numeric features are computed from the raw tweet text so that capitalization and punctuation cues are preserved.

Group	Count	Description
TF-IDF	8,000	Unigrams to trigrams; min_df = 2, max_df = 0.95, sublinear TF, unicode accent stripping
Linguistic	6	Text length, word count, average word length, '!' count, '?' count, capital-letter ratio
Sentiment	4	VADER positive, negative, neutral and compound scores
Offensive	1	Count of matches against a small offensive-word list
Total	8,011	Numeric features scaled to [0, 1] with MinMaxScaler, then concatenated with TF-IDF

Table 3. Feature groups.

With sublinear term frequency, the TF-IDF weight of term t in document d is:

$$w(t, d) = (1 + \ln tf(t, d)) \times idf(t), \quad idf(t) = \ln[(1 + n) / (1 + df(t))] + 1 \qquad (1)$$

where tf(t, d) is the count of t in d, df(t) is the number of documents containing t and n is the number of documents. Min–max scaling maps each numeric feature x to $x' = (x - x_min) / (x_max - x_min)$, which also keeps the values non-negative as required by Multinomial Naïve Bayes.

4.4 Classifiers

Classifier	Configuration	Rationale
SVM (LinearSVC)	C = 1.0, max_iter = 2000	Effective for high-dimensional sparse text
Logistic Regression	C = 1.0, max_iter = 1000	Strong, interpretable baseline
Random Forest	100 trees, random_state = 42	Captures non-linear relationships
Multinomial Naïve Bayes	alpha = 0.1	Efficient probabilistic text model
KNN	k = 5	Similarity-based; distance-driven

Table 4. Base classifiers.

4.5 Hard-Voting Ensemble

Let $h_1, \ldots, h_5$ be the five classifiers, each predicting a class in {0, 1}. The hard-voting ensemble outputs the majority class:

$$\hat{y}(x) = \arg\max_c \sum_{i=1}^{5} 1[h_i(x) = c], \quad c \in \{0, 1\} \tag{2}$$

Because five voters and two classes can never tie, no tie-breaking rule is needed. For example, if SVM, LR and NB predict Cyberbullying while RF and KNN predict Not Cyberbullying, the final output is Cyberbullying (3 of 5 votes). No meta-learner is trained.

4.6 Data Splitting and Training

The 17,805 tweets were split with stratification (random_state = 42) into 70% training (12,463), 15% validation (2,671) and 15% test (2,671) samples. The validation set was used to compare models, and the test set was reserved for final evaluation. All classifiers were trained on the same training samples, and the ensemble was additionally evaluated with 5-fold cross-validation on the training set. Implementation uses Python, scikit-learn, NLTK (VADER), pandas, Matplotlib and Seaborn.

Subset	Samples	Share (%)
Training	12,463	70.0
Validation	2,671	15.0
Test	2,671	15.0
Total	17,805	100

Table 5. Stratified data split.

4.7 Evaluation Metrics

With TP, TN, FP and FN denoting true positives, true negatives, false positives and false negatives (cyberbullying = positive class), the metrics are:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \tag{3}$$

$$Precision = TP / (TP + FP), \quad Recall = TP / (TP + FN) \tag{4}$$

$$F1 = 2 \times Precision \times Recall / (Precision + Recall) \tag{5}$$

Cross-validation accuracy is summarized as the mean $\bar{x}$ and standard deviation σ over the k = 5 folds.

5. Results

5.1 Ensemble Performance on the Test Set

The ensemble achieved 83.08% accuracy on the unseen test set. Per-class results are given in Table 6 and Figure 4(a).

Class	Precision	Recall	F1-score	Support
Not Cyberbullying	0.78	0.86	0.82	1,171
Cyberbullying	0.88	0.81	0.84	1,500
Macro average	0.83	0.83	0.83	2,671
Overall accuracy			0.83	2,671

Table 6. Classification report of the ensemble (test set). Macro averages computed from the per-class values.

5.2 Cross-Validation

Five-fold cross-validation on the training set gave fold accuracies of 82.15%, 81.87%, 83.91%, 83.31% and 82.50%. The mean is $\bar{x} = 82.75\%$ with a fold standard deviation of $\sigma \approx 0.76$ points (the reported $\pm$ 1.51% corresponds to 2σ). The CV mean is within 0.4 points of the test accuracy, which indicates stable behavior and no sign of overfitting to a particular split (Figure 4(b)).

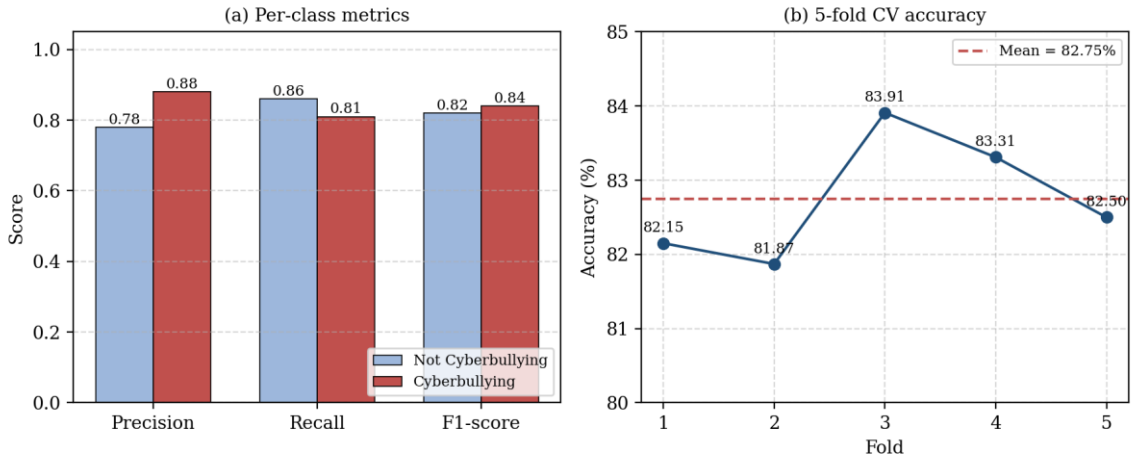


Figure 4. (a) Per-class precision, recall and F1-score; (b) fold-wise cross-validation accuracy.

5.3 Confusion Matrix

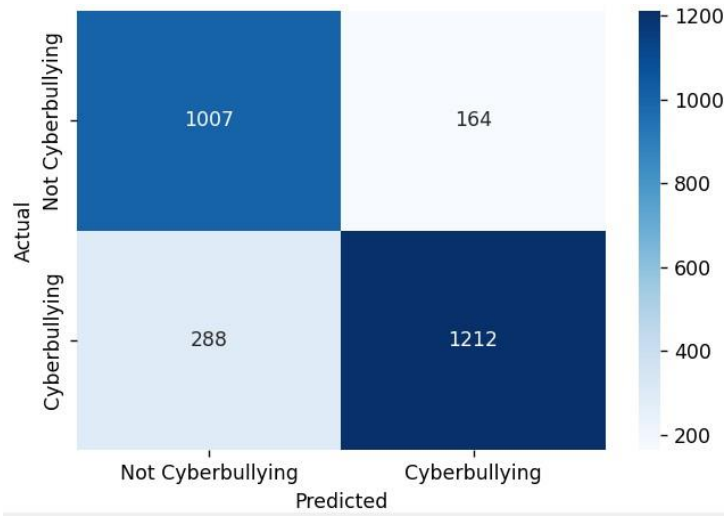


Figure 5. Confusion matrix of the ensemble on the test set.

The ensemble correctly cleared 1,007 non-cyberbullying tweets (TN) and detected 1,212 cyberbullying tweets (TP), with 164 false alarms (FP) and 288 missed cases (FN). Accuracy = (1,007 + 1,212) / 2,671 = 83.08%. Additional statistics derived from the confusion matrix are listed in Table 7.

Statistic	Computation	Value
Specificity (TNR)	1,007 / (1,007 + 164)	0.860
False-positive rate	164 / 1,171	0.140
False-negative rate (miss rate)	288 / 1,500	0.192
Negative predictive value	1,007 / (1,007 + 288)	0.778
Balanced accuracy	(0.808 + 0.860) / 2	0.834

Table 7. Statistics derived from the confusion matrix.

Precision for the cyberbullying class (0.88) exceeds its recall (0.81): the ensemble is conservative, raising few false alarms but missing about 19% of harmful tweets. In a moderation setting, missed cyberbullying is the costlier error, so reducing the 288 false negatives is the main priority.

5.4 Comparison with Individual Models

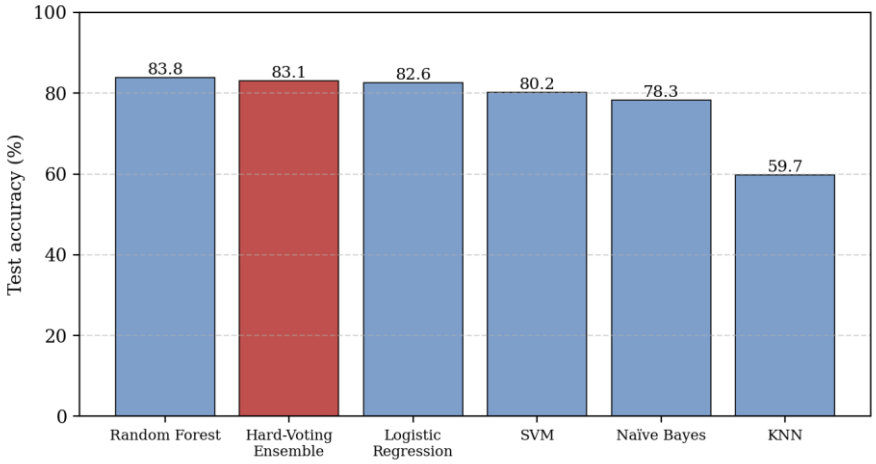


Figure 6. Test accuracy of the individual classifiers and the hard-voting ensemble (individual values read from the project's accuracy chart; approximate).

Model	Test accuracy (%)
Random Forest	≈ 83.8
Hard-Voting Ensemble	83.08
Logistic Regression	≈ 82.6
SVM (LinearSVC)	≈ 80.2
Naïve Bayes	≈ 78.3
KNN	≈ 59.7

Table 8. Test accuracy by model.

6. Discussion

- **Competitive and stable.** The ensemble reaches 83.08% test and 82.75% CV accuracy with consistent per-class behavior.
- **Not better than the best single model.** It is about 0.7 points below Random Forest, so the research question is only partly confirmed: the ensemble is reliable, but not strictly more accurate than every member. Its value is robustness—no single best classifier has to be chosen in advance.
- **Robust to weak voters.** KNN (≈ 59.7%) degrades because distance measures lose meaning in a sparse 8,011-dimensional space, yet majority voting keeps the ensemble roughly 23 points above it.
- **Comparison with the literature.** The 83.08% accuracy is lower than the 90–96% reported in Table 1. This should not be read as a like-for-like gap: those studies use different datasets, preprocessing and protocols, and our task merges five heterogeneous abuse categories (religion, age, gender, ethnicity, other) into one class, which makes the binary decision harder.

- **Error analysis.** Sarcasm, indirect insults and context-dependent abuse are difficult for TF-IDF-based models and probably account for many of the 288 false negatives. Offensive-word counting is keyword-based and cannot capture such cases.

7. Limitations

- Only one English Twitter dataset was used, so generalization to other platforms or languages is untested.
- No feature ablation was performed, so the individual contribution of the linguistic and sentiment features is unknown.
- Hyperparameters were left at default or fixed values and were not tuned.
- The offensive-word list is small and keyword-based; KNN is both weak and slow.
- The TF-IDF vectorizer and scaler were fitted on the whole balanced dataset before splitting, which may slightly inflate reported scores.
- Individual model accuracies in Figure 6 are read from the project's chart and are approximate.

8. Future Work

- Soft or weighted voting and stacking; removal of KNN.
- Hyperparameter tuning with grid or random search.
- Recall-oriented threshold tuning to reduce false negatives.
- Fitting feature extractors on the training split only; feature ablation study.
- Deep models such as LSTM or BERT, and deployment as a web application.

9. Conclusion

A hard-voting ensemble of five traditional classifiers over 8,011 TF-IDF, linguistic and sentiment features detects cyberbullying in tweets with 83.08% test accuracy and an F1-score of 0.84 for the cyberbullying class. It outperforms SVM, Naïve Bayes and KNN, matches Logistic Regression and trails Random Forest by under one point, while remaining robust to weak members. The results support the ensemble as a reliable, simple detector rather than a clear accuracy improvement over the best single model. Reducing missed cyberbullying cases is the main priority for further improvement.

References

[1] K. S. Alam, S. Bhowmik, and P. R. K. Prosun, "Cyberbullying Detection: An Ensemble Based Machine Learning Approach," ICICV 2021, pp. 710–715.

[2] A. F. Alqahtani and M. Ilyas, "A Machine Learning Ensemble Model for the Detection of Cyberbullying," Int. J. Artificial Intelligence & Applications, vol. 15, no. 1, pp. 115–129, 2024.

[3] M. Singh, A. Metta, and S. Patnaik, "Optimizing Cyberbullying Detection in Tweets with an Ensemble Learning Framework," J. Informatics Education and Research, vol. 5, no. 3, 2025.

[4] andrewmvd, "Cyberbullying Classification," Kaggle dataset (cyberbullying_tweets.csv).

[5] C. J. Hutto and E. Gilbert, "VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text," ICWSM, 2014.